

5.4 #5 Let $\phi(x) = \begin{cases} 0 & \text{for } 0 < x < 1 \\ 1 & \text{for } 1 < x < 3 \end{cases}$

$$(a) A_0 = \frac{2}{l} \int_0^l \phi(x) dx = \frac{2}{3} \int_0^3 \phi(x) dx = \frac{4}{3}$$

$$(m \neq 0), A_m = \frac{2}{l} \int_0^l \phi(x) \cos\left(\frac{m\pi x}{l}\right) dx$$

$$(l=3) = \frac{2}{3} \int_0^3 \phi(x) \cos\left(\frac{m\pi x}{3}\right) dx$$

$$= \frac{2}{3} \int_1^3 \cos\left(\frac{m\pi x}{3}\right) dx = \frac{2}{3} \cdot \frac{3}{m\pi} \sin\left(\frac{m\pi x}{3}\right) \Big|_1^3$$

$$= \frac{2}{m\pi} [\sin(m\pi) - \sin\left(\frac{m\pi}{3}\right)] = -\frac{2}{m\pi} \sin\left(\frac{m\pi}{3}\right)$$

Fourier cosine series:

$$S := \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cdot \cos\left(\frac{m\pi x}{l}\right)$$

$$= \frac{2}{3} + \sum_{m=1}^{\infty} \left(-\frac{2}{m\pi}\right) \sin\left(\frac{m\pi}{3}\right) \cdot \cos\left(\frac{m\pi x}{3}\right)$$

$$= \frac{2}{3} + -\frac{2}{\pi} \cdot \sin\left(\frac{\pi}{3}\right) \cdot \cos\left(\frac{\pi x}{3}\right) + -\frac{2}{2\pi} \cdot \sin\left(\frac{2\pi}{3}\right) \cdot \cos\left(\frac{2\pi x}{3}\right)$$

$$+ -\frac{2}{4\pi} \cdot \sin\left(\frac{4\pi}{3}\right) \cdot \cos\left(\frac{4\pi x}{3}\right) + \dots$$

$$= \frac{2}{3} + -\frac{\sqrt{3}}{\pi} \cdot \cos\left(\frac{\pi x}{3}\right) + -\frac{\sqrt{3}}{2\pi} \cdot \cos\left(\frac{2\pi x}{3}\right) + \frac{\sqrt{3}}{4\pi} \cos\left(\frac{4\pi x}{3}\right) + \dots$$

(b) By Strauss Chp 5.4, Thm 4.8.

$\phi(x)$ and $\phi'(x)$ are piecewise continuous (by extending ϕ to an even function $\tilde{\phi}$ on $(-l, l)$), thus

$$\frac{2}{3} + \sum_{m=1}^{\infty} \left(-\frac{2}{m\pi}\right) \cdot \sin\left(\frac{m\pi}{3}\right) \cdot \cos\left(\frac{m\pi x}{3}\right) = \begin{cases} 0 & 0 \leq x < 1 \\ \frac{\phi(1^+) + \phi(1^-)}{2} = \frac{1}{2}, & x=1 \\ 1 & 1 < x \leq 3. \end{cases}$$

$$(c) \int_{-\infty}^{\infty} |\phi(x)|^2 = \int_1^3 1^2 = 2 < \infty$$

$\Rightarrow \phi(x) \in L^2 \Rightarrow$ S converge to $\phi(x)$ in L^2 sense
By Strauss Chp 5.4, Thm 3.

$$\begin{aligned} (d) 0 = \phi(0) &= \frac{2}{3} + \sum_{m=1}^{\infty} \left(-\frac{2}{m\pi}\right) \cdot \sin\left(\frac{m\pi}{3}\right) \cdot \cos\left(\frac{m\pi \cdot 0}{3}\right) \\ &= \frac{2}{3} + -\frac{\sqrt{3}}{\pi} \cdot \cos\left(\frac{\pi \cdot 0}{3}\right) + -\frac{\sqrt{3}}{2\pi} \cdot \cos\left(\frac{2\pi \cdot 0}{3}\right) + \frac{\sqrt{3}}{4\pi} \cos\left(\frac{4\pi \cdot 0}{3}\right) + \dots \\ &= \frac{2}{3} + -\frac{\sqrt{3}}{\pi} - \frac{\sqrt{3}}{2\pi} + \frac{\sqrt{3}}{4\pi} + \dots = \frac{2}{3} + \left(-\frac{\sqrt{3}}{\pi}\right) \cdot \left(1 + \frac{1}{2} - \frac{1}{4} - \frac{1}{5} + \dots\right) \end{aligned}$$

$$\Rightarrow 1 - \frac{1}{2} - \frac{1}{4} - \frac{1}{5} + \dots = \frac{2\pi}{3\sqrt{3}}$$

S.5 #13 (a) f is a C^1 fcn of period 2π .

By translation we may assume $f(0)=0$.

(i.e. consider $\tilde{f} \stackrel{\text{def}}{=} f(x) - f(0)$),

Thus it suffices to show $S_N(x)$ converges to 0 at $x=0$ as $N \rightarrow \infty$ where

$$S_N(x) \stackrel{\text{def}}{=} \frac{A_0}{2} + \sum_{n=1}^N A_n \cos\left(\frac{n\pi x}{l}\right) + B_n \sin\left(\frac{n\pi x}{l}\right) \quad (l=\pi)$$

is a partial sum $n=1$

(b) $g(x) = \frac{f(x)}{e^{ix}-1}$ is always continuous if $e^{ix}-1 \neq 0$.

By periodicity of $g(x)$, i.e. $g(x+2\pi) = g(x)$, we may always discuss over $x \in [-\pi, \pi]$.

Note $e^{ix}-1=0 \Leftrightarrow \cos x + i \sin x - 1 = 0 \Leftrightarrow x=0$

By L'Hopital's rule,

$$\lim_{x \rightarrow 0} g(x) \stackrel{\text{(L'H)}}{=} \frac{\lim_{x \rightarrow 0} f'(x)}{\lim_{x \rightarrow 0} (e^{ix}-1)'} = \frac{f'(0)}{ie^{i \cdot 0}} = \frac{f'(0)}{i}, \quad f \in C^1.$$

NB: I believe the author really means:

Show that g has a continuous extension, given by

$$\tilde{g}(x) \stackrel{\text{def}}{=} \begin{cases} g(x) & x \neq 2k\pi, k \in \mathbb{Z} \\ \frac{f'(0)}{i} & x = 2k\pi, k \in \mathbb{Z}. \end{cases}$$

such that $\tilde{g}$ is continuous over $\mathbb{R}$.

(c) $D_n \stackrel{\text{def}}{=} \text{the coefficients of } \tilde{g}(x)$. Show that $D_n \rightarrow 0$.
 Proof: Since $\tilde{g}(x) \in C([- \pi, \pi])$, $\tilde{g} \in L^1([- \pi, \pi])$.

By Riemann-Lebesgue, $\lim_{n \rightarrow \pm \infty} D_n = 0$.

(Unfortunately, I didn't see a proof without Riemann-Lebesgue. The original paper also uses it.)

(d) We simply compute

$$D_{n-1} - D_n = \frac{1}{2\pi} \left(\int_{-\pi}^{\pi} g(x) e^{-i(n-1)x} dx - \int_{-\pi}^{\pi} g(x) e^{-inx} dx \right)$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} g(x) e^{-inx} (e^{ix} - 1) dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \quad (\text{by def'n of } g).$$

$$= C_n \quad (\text{by def'n of } C_n)$$

(e) Thus $\sum_{n=-N}^N C_n = \sum_{n=-N}^N (D_{n-1} - D_n) = D_{N-1} - D_N \quad (*)$

Letting $N \rightarrow +\infty$ in $(*)$ and note $\lim_{N \rightarrow \pm \infty} D_N = 0$ by (c).

we have $\sum_{n=-\infty}^{\infty} C_n = 0 - 0 = 0$

i.e., $\sum_{n=-\infty}^{\infty} C_n e^{in \cdot 0} = f(0)$, which suffices by (a).

$$4.1\#4 \quad \begin{cases} u_{tt} = c^2 u_{xx} - \gamma u_t & 0 < x < l & (1) \\ u(0, t) = u(l, t) = 0 & & (2) \\ u(x, 0) = \phi(x) & & (3) \\ u_t(x, 0) = \psi(x) & & (4) \end{cases}$$

where $0 < \gamma < 2\pi c/l$ is a constant.

Write down the series expansion of the solution.

Proof: $u(x, t) = X(x)T(t)$.

By (1), $X T_{tt} = c^2 X_{xx} T - \gamma X T_t$

$$\Rightarrow T_{tt} = c^2 \frac{X_{xx} T}{X} - \gamma T_t$$

$$\Rightarrow \frac{X_{xx}}{X} = \frac{T_{tt} + \gamma T_t}{c^2 T} \stackrel{\text{def}}{=} k$$

By (2), $X(0)T(t) = X(l)T(t) = 0 \Rightarrow X(0) = X(l) = 0$.

(i) $k = \lambda^2 > 0$ (Positive eigenval) $\cosh(x) := \frac{e^x + e^{-x}}{2}$
 $\sinh(x) = \frac{e^x - e^{-x}}{2}$

$$X = A \cosh(\lambda x) + B \sinh(\lambda x)$$

$$X(0) = 0 \Rightarrow 0 = A \cosh(0) + B \sinh(0) = A$$

$$X(l) = 0 \Rightarrow 0 = A \cosh(\lambda l) + B \sinh(\lambda l) \Rightarrow B = 0$$

$\Rightarrow k$ cannot be positive

(ii) $k = 0$, $X_{xx} = 0 \Rightarrow X = Ax + B$

$$X(0) = 0 \Rightarrow X(0) = A \cdot 0 + B \Rightarrow B = 0$$

$$X(l) = 0 \Rightarrow 0 = A \cdot l + B \Rightarrow A = 0 \text{ (since } l > 0 \text{)}.$$

$\Rightarrow k$ cannot be zero.

(iii) $k = -\lambda^2 < 0$. (negative eigenval)

$$X = A \cos(\lambda x) + B \sin(\lambda x)$$

$$X(0) = 0 \Rightarrow X(0) = A \cdot \cos(\lambda \cdot 0) + B \cdot \sin(\lambda \cdot 0) = A$$

$$X(l) = 0 \Rightarrow 0 = A \cos(\lambda l) + B \sin(\lambda l) = B \sin(\lambda l).$$

If $B \neq 0$, then $\sin(\lambda l) = 0 \Rightarrow \lambda l = n\pi, n \in \mathbb{Z}^+$

$$\text{Eigenvalue } \lambda_n = \frac{n\pi}{l} \Rightarrow X_n(x) = \sin\left(\frac{n\pi}{l}x\right).$$

We solve for T :

$$\frac{T_{tt} + rT_t}{c^2 T} = k = -\lambda^2$$

$$\Rightarrow T_{tt} + rT_t + \lambda^2 c^2 T = 0$$

The characteristic polynomial is

$$X^2 + rX + \lambda^2 c^2 = 0.$$

$$\Rightarrow X_{1,2} = \frac{-r \pm \sqrt{r^2 - 4\lambda^2 c^2}}{2}.$$

Note $0 < r < 2\pi c/l$, so $r^2 < \left(\frac{2\pi c}{l}\right)^2 = \frac{4\pi^2}{l^2} \cdot c^2 < 4\lambda_n^2 c^2$

for any n , since $\lambda_n = \frac{n\pi}{l} \Rightarrow r^2 - 4\lambda_n^2 c^2 < 0$.

$$\Rightarrow X_{1,2} = \frac{-r \pm i\sqrt{4\lambda^2 c^2 - r^2}}{2}$$

$$\Rightarrow T(t) = e^{-\frac{r}{2}t} \left(A \cos\left(\frac{\sqrt{4\lambda^2 c^2 - r^2}}{2} t\right) + B \sin\left(\frac{\sqrt{4\lambda^2 c^2 - r^2}}{2} t\right) \right)$$

$$u(x,t) = \sum_{n=1}^{\infty} X_n(x) \cdot T_n(t)$$

$$= \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{\ell}x\right) e^{-\frac{r}{2}t} \left(A_n \cos\left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}t\right) + B_n \sin\left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}t\right) \right) \quad (*)$$

A_n, B_n to be determined, $\lambda_n \stackrel{\text{def}}{=} \frac{n\pi}{\ell}$.

By (3), $u(x,0) = \phi(x)$, plugging $t=0$ into (*), then

$$\phi(x) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{\ell}x\right) \cdot A_n$$

$$\Rightarrow A_n = \frac{2}{\ell} \int_0^{\ell} \phi(x) \sin\left(\frac{n\pi x}{\ell}\right) dx \text{ by uniqueness of Fourier series}$$

By (4), $u_t(x,0) = \psi(x)$

$$u_t(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \left[\left(-\frac{r}{2} e^{-\frac{r}{2}t}\right) \left(A_n \cos\left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}t\right) + B_n \sin\left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}t\right) \right) \right. \\ \left. + e^{-\frac{r}{2}t} \cdot \left(-A_n \sin\left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}t\right) \cdot \left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}\right) + B_n \cos\left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}t\right) \cdot \left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}\right) \right) \right]$$

Evaluating at $t=0$, then

$$\psi(x) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \left[\left(-\frac{r}{2} A_n\right) + B_n \cdot \left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}\right) \right]$$

$$\Rightarrow -\frac{r}{2} A_n + B_n \left(\frac{\sqrt{4\lambda_n^2 c^2 - r^2}}{2}\right) = \frac{2}{\ell} \int_0^{\ell} \psi(x) \sin\left(\frac{n\pi x}{\ell}\right) dx$$

by uniqueness of Fourier series

$$B_n = \frac{2}{\sqrt{4\lambda_n^2 c^2 - r^2}} \cdot \left(\frac{r}{2} A_n + \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{n\pi x}{l}\right) dx \right)$$

$$= \frac{2}{\sqrt{4 \cdot \left(\frac{n\pi}{l}\right)^2 c^2 - r^2}} \cdot \frac{2}{l} \int_0^l \left(\frac{r}{2} \phi + \psi \right) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{4}{\sqrt{4n^2\pi^2 c^2 - r^2 l^2}} \int_0^l \left(\frac{r}{2} \phi + \psi \right) \sin\left(\frac{n\pi x}{l}\right) dx$$

4.1 #6 Separate the variables for

$$\begin{cases} t u_t - u_{xx} = 2u & (1) \\ u(0,t) = u(\pi,t) = 0 & (2) \end{cases}$$

Show that there are ∞ number of solutions s.t.
 $u(x,0) = 0.$ (3)

Proof: $u(x,t) = X(x)T(t)$

By (1), $t X T_t - X_{xx} T = 2 X \cdot T$

$$\frac{t \overline{T}_t}{\overline{T}} \cdot X - X_{xx} = 2X$$

$$\frac{t \cdot \overline{T}_t}{\overline{T}} = \frac{X_{xx} + 2X}{X} = \frac{X_{xx}}{X} + 2$$

$$\Rightarrow \frac{t \cdot \overline{T}_t}{\overline{T}} - 2 = \frac{X_{xx}}{X} \stackrel{\text{def}}{=} k$$

By (2), $X(0)T(t) = X(\pi)T(t) = 0$
 $\Rightarrow X(0) = X(\pi) = 0.$

(i) $k = \lambda^2 > 0$ (Positive eigenval) $\sinh(x) = \frac{e^{x^2} - e^{-x}}{2}.$

$$X = A \cosh(\lambda x) + B \sinh(\lambda x)$$

$$X(0) = 0 \Rightarrow 0 = A \cosh(0) + B \sinh(0) = A$$

$$X(\pi) = 0 \Rightarrow 0 = A \cosh(\lambda \pi) + B \sinh(\lambda \pi) \Rightarrow B = 0$$

$\Rightarrow k$ cannot be positive

$$(ii) k=0, X_{xx}=0 \Rightarrow X=Ax+B$$

$$X(0)=0 \Rightarrow X(0)=A \cdot 0 + B \Rightarrow B=0$$

$$X(\pi)=0 \Rightarrow 0=A \cdot \pi + B \Rightarrow A=0 \text{ (since } \pi > 0 \text{)}.$$

$\Rightarrow k$ cannot be zero.

$$(iii) k = -\lambda^2 < 0. \text{ (negative eigenval)}$$

$$X = A \cos(\lambda x) + B \sin(\lambda x)$$

$$X(0)=0 \Rightarrow X(0)=A \cdot \cos(\lambda \cdot 0) + B \cdot \sin(\lambda \cdot 0) = A$$

$$X(\pi)=0 \Rightarrow 0 = A \cos(\lambda \pi) + B \sin(\lambda \pi) = B \sin(\lambda \pi).$$

If $B \neq 0$, then $\sin(\lambda \pi) = 0 \Rightarrow \lambda \pi = n\pi, n \in \mathbb{Z}^+$

$$\text{Eigenvalue } \lambda_n = \frac{n\pi}{\pi} = n \Rightarrow X_n(x) = \sin(nX)$$

$$\text{We solve for } T: \frac{t \cdot T_t}{1} - 2 = -n^2$$

$$T_t - \frac{1}{t}(2-n^2)T = 0$$

$$\Rightarrow \mu(t) = e^{\int_0^t -\frac{1}{t}(2-n^2)} = e^{(n^2-2)\ln t} = t^{n^2-2}$$

$$\Rightarrow T = \frac{C}{\mu(t)} = \frac{C}{t^{n^2-2}} = C t^{2-n^2}.$$

$$\Rightarrow T_n(t) = t^{2-n^2}.$$

$$\text{Thus } u(x,t) = \sum_{n=1}^{\infty} A_n X_n(x) T_n(t)$$

$$= \sum_{n=1}^{\infty} A_n \cdot \sin(nX) \cdot t^{2-n^2}. \quad A_n \text{ TBD.}$$

By (3), $0 = u(x, 0) = \sum_{n=1}^{\infty} A_n \sin(nx) \cdot 0^{2-n^2} = 0$.
We can choose any A_n .

4.2 #2 Consider
$$\begin{cases} u_{tt} = c^2 u_{xx} & (1) \quad 0 < x < l \\ u_x(0, t) = 0 & (2) \\ u(l, t) = 0 & (3) \end{cases}$$

(a) Show that the eigenfunctions are $\cos[(n + \frac{1}{2})\pi x / l]$

Say $u(x, t) = X(x)T(t)$, then by (1),

$$X T_{tt} = c^2 X_{xx} T$$

$$\Rightarrow \frac{X_{xx}}{X} = \frac{1}{c^2} \cdot \frac{T_{tt}}{T} \stackrel{\text{def}}{=} k$$

By (2), $X_x(0) \cdot T(t) = 0 \Rightarrow X_x(0) = 0$
By (3), $X(l) \cdot T(t) = 0 \Rightarrow X(l) = 0$

(i) $k = \lambda^2 > 0$ (Positive eigenval) $\sinh(x) = \frac{e^x - e^{-x}}{2}$

$$X = A \cosh(\lambda x) + B \sinh(\lambda x)$$

$$X_x(0) = 0 \Rightarrow 0 = \lambda(A \sinh(0) + B \cosh(0)) = B$$

$$X(l) = 0 \Rightarrow 0 = A \cosh(\lambda l) + B \sinh(\lambda l) \Rightarrow A = 0$$

$\Rightarrow k$ cannot be positive

(ii) $k = 0$, $X_{xx} = 0 \Rightarrow X = Ax + B$

$$X_x(0) = 0 \Rightarrow 0 = A \Rightarrow A = 0$$

$$X(l)=0 \Rightarrow 0=0 \cdot l + B \Rightarrow B=0$$

$\Rightarrow k$ cannot be zero.

(iii) $k = -\lambda^2 < 0$. (negative eigenval)

$$X = A \cos(\lambda x) + B \sin(\lambda x)$$

$$X(0)=0 \Rightarrow 0 = -A\lambda \sin(\lambda \cdot 0) + B\lambda \cos(\lambda \cdot 0) = B\lambda \Rightarrow B=0$$

$$X(l)=0 \Rightarrow 0 = A \cos(\lambda l) + B \cdot \sin(\lambda l) = A \cos(\lambda l)$$

$$A \neq 0, \text{ so } \cos(\lambda l) = 0 \Rightarrow \lambda l = n\pi + \frac{\pi}{2}, n \in \mathbb{N}.$$

$$\Rightarrow \lambda_n = \pi(n + \frac{1}{2}) \cdot \frac{1}{l}.$$

The corresponding X_n is then

$$X_n = \cos[(n + \frac{1}{2})\pi x / l].$$

(b) Write the series expansion for $u(x,t)$

$$\text{Once we know } X_n = \cos[(n + \frac{1}{2})\pi x / l],$$

$$\text{with } k = -\lambda_n^2 = -(\pi(n + \frac{1}{2}) \frac{1}{l})^2$$

$$\text{we have } \frac{1}{c^2} \cdot \frac{T_{tt}}{T} = k = -\lambda_n^2$$

$$\Rightarrow T_n(t) = A_n \cos(c\lambda_n t) + B_n \sin(c\lambda_n t)$$

$$\Rightarrow u(x,t) = \sum_{n=0}^{\infty} X_n \cdot T_n$$

$$= \sum_{n=0}^{\infty} (\cos[(n + \frac{1}{2})\pi x / l]) [A_n \cos(c\lambda_n t) + B_n \sin(c\lambda_n t)]$$

where $\lambda_n = \pi(n + \frac{1}{2}) \frac{1}{l}$. A_n, B_n TBD.